

Sensor HW6

April 26, 2026

1.

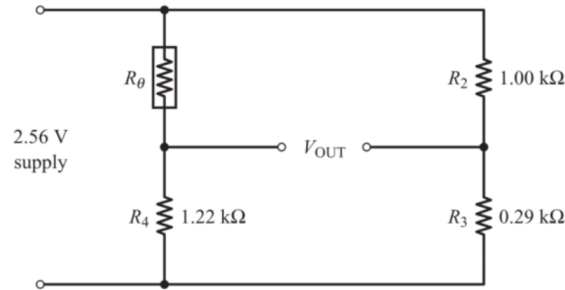
The resistance R_θ k Ω of a thermistor at θ K is given by:

$$R_\theta = 1.68 \exp \left[3050 \left(\frac{1}{\theta} - \frac{1}{298} \right) \right]$$

The thermistor is incorporated into the deflection bridge circuit shown in Fig. Prob. 3.

- (a) Assuming that V_{OUT} is measured with a detector of infinite impedance, calculate:
- the range of V_{OUT} corresponding to an input temperature range of 0 to 50 °C;
 - the non-linearity at 12 °C as a percentage of full-scale deflection.
- (b) Calculate the effect on the range of V_{OUT} of reducing the detector impedance to 1 k Ω .

Figure Prob. 3.



(a)

When temperature is 0 °C, $\theta = 273K$

$$R_{\theta@273} = 1.68 \exp \left[3050 \left(\frac{1}{273} - \frac{1}{298} \right) \right] = 4.289 K\Omega$$

$$V_{right} = V_s \frac{R_3}{R_2 + R_3} = 0.5755[V], V_{left} = \frac{R_4}{R_\theta + R_4} = 0.5669[V]$$

$$V_{out} = V_{left} - V_{right} = -0.0086[V]$$

When temperature is 50 °C, $\theta = 273 + 50 = 323K$.

$$R_{\theta@323} = 1.68 \exp \left[3050 \left(\frac{1}{273} - \frac{1}{323} \right) \right] = 0.76 K\Omega$$

$$V_{right} = V_s \frac{R_3}{R_2 + R_3} = 0.5755[V], V_{left} = \frac{R_4}{R_\theta + R_4} = 1.5767[V]$$

$$V_{out} = V_{left} - V_{right} = 1.001[V]$$

Range of V_{out} from -0.0086V to 1.001V

(ii) When temperature is 12 °C, $\theta = 273 + 12 = 285K$

$$R_{\theta@285} = 1.68 \exp \left[3050 \left(\frac{1}{273} - \frac{1}{285} \right) \right] = 2.68 K\Omega$$

$$V_{right} = V_s \frac{R_3}{R_2 + R_3} = 0.5755[V], V_{left} = \frac{R_4}{R_\theta + R_4} = 0.8009[V]$$

$$V_{out} = V_{left} - V_{right} = 0.2254[V]$$

Assume R_θ increases linearly with temperature.

$$V_{\theta ideal} = V_{out@285} + \frac{V_{out@323} - V_{out@273}}{50} 12 = 0.2337[V]$$

$$error = V_{out} - V_{ideal} = -0.0084$$

$$\text{non-linearity : \%f.s.d} = \frac{-0.0084}{V_{out@323} - V_{out@273}} \times 100\% = \boxed{-0.83\%}$$

(b)

From Thévenin equivalent circuit :

$$R_{th} = (R_\theta \parallel R_4) + (R_2 \parallel R_3) = \frac{R_\theta R_4}{R_\theta + R_4} + \frac{R_2 R_3}{R_2 + R_3} = 0.2248 + \frac{1.22 R_\theta}{1.22 + R_\theta}$$

$$R_{th@273} = 0.2248 + \frac{1.22 \cdot 4.289}{1.22 + 4.289} = 1.1746\Omega$$

$$R_{th@323} = 0.2248 + \frac{1.22 \cdot 0.76}{1.22 + 0.76} = 0.693\Omega$$

According to Voltage division law:

$$V'_{out} = V_{out} \cdot \frac{R_d}{R_d + R_{th}} = V_{out} \cdot \frac{1}{1 + R_{th}}$$

$$V'_{out@273} = V_{out@273} \cdot \frac{1}{1 + 1.1746} = -0.00395[V]$$

$$V'_{out@323} = V_{out@323} \cdot \frac{1}{1 + 0.693} = 0.591[V]$$

Range of V'_{out} from $\boxed{-0.00395V \text{ to } 0.5912V}$

2.

A load cell consists of a domed vertical steel cylinder 20 cm high and 15 cm in diameter. Four flat surfaces, at right angles to each other, are cut on the vertical surface so as to form 10 cm squares. Resistance strain gauges are attached to these flat surfaces so that two gauges (on opposite faces) suffer longitudinal compression and two gauges (on the other pair of opposite faces) suffer transverse tension. The strain gauges have the following specification:

Resistance	= 100 Ω
Gauge factor	= 2.1
Maximum gauge current	= 30 mA

The gauges are connected in a temperature-compensated bridge and the out-of-balance signal is input to a differential amplifier. Calculate the minimum amplifier gain if the amplifier output voltage is to be 1 V for a compressive force of 10^5 N.

Young's modulus for steel	= 2.1×10^{11} N m ⁻²
Poisson's ratio for steel	= 0.29

$$\epsilon_L = \frac{\sigma}{E} = \frac{F/A}{E} = \frac{-10^5/\pi \cdot 0.075^2}{2.1 \cdot 10^{11}} = 2.69 \cdot 10^{-5}$$

$$\Delta R = RG\epsilon_L = 100 \cdot 2.1 \cdot 2.69 \cdot 10^{-5} = 0.00566\Omega$$

$$R_{max} = R + \Delta R = 100.00566\Omega$$

$$\epsilon_T = -v\epsilon_L = 0.29 \cdot 2.69 \cdot 10^{-5} = 0.78 \cdot 10^{-5}$$

$$\Delta R = RG\epsilon_T = 100 \cdot 2.1 \cdot 0.78 \cdot 10^{-5} = 0.000164\Omega$$

$$R_{min} = R - \Delta R = 99.9984\Omega$$

$$E_{th} = \left(\frac{R_{min}}{R_{min} + R_{max}} - \frac{R_{max}}{R_{min} + R_{max}} \right) \cdot V_s = -0.0001824[V]$$

$$\mathbf{gain} = \frac{V_{amp,out}}{V_{out}} = \frac{1}{E_{th}} = \boxed{-5481}$$